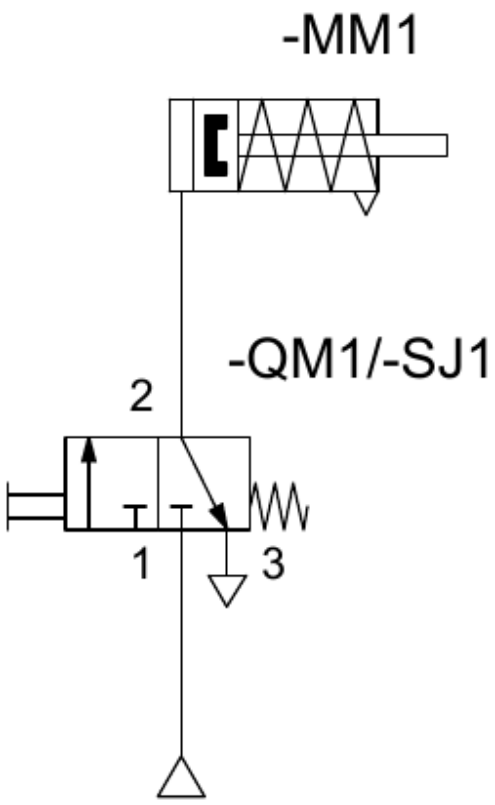
MES0085 — PN-Lab Final Circuits

All 9 FluidSIM exercise circuits, built and verified by Jordan Brodie ahead of the Wed 23.09.2026 lab (group 5 Pneumo).

Exercise 1 — Single-acting cylinder, basic control

Suggested filename: Ex1-SingleActing-Basic.circ

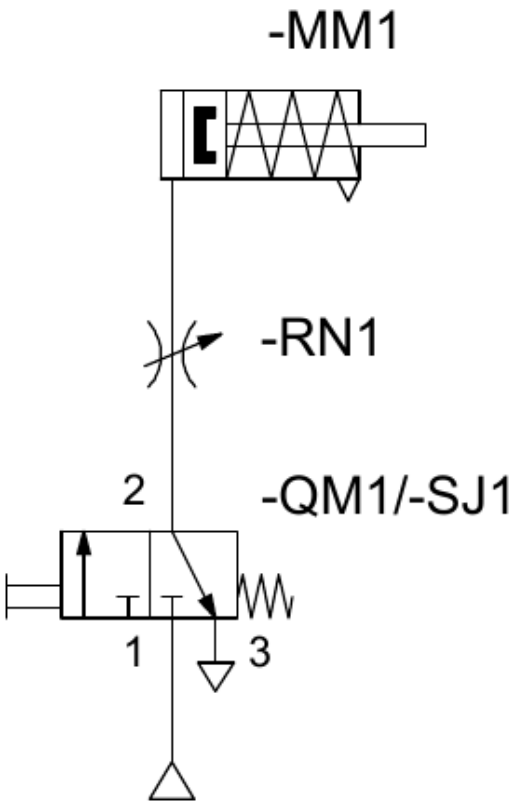
A single-acting, spring-return cylinder (-MM1) is controlled by a 3/2-way valve with pushbutton actuator, normally closed (-QM1/-SJ1). Pressing the button connects the supply (port 1) to the cylinder (port 2), extending it; releasing the button lets the spring return the valve, venting the cylinder through port 3 so the cylinder's own return spring retracts it.



Exercise 2 — Single-acting cylinder, adjustable speed (one direction)

Suggested filename: Ex2-SingleActing-Throttle.circ

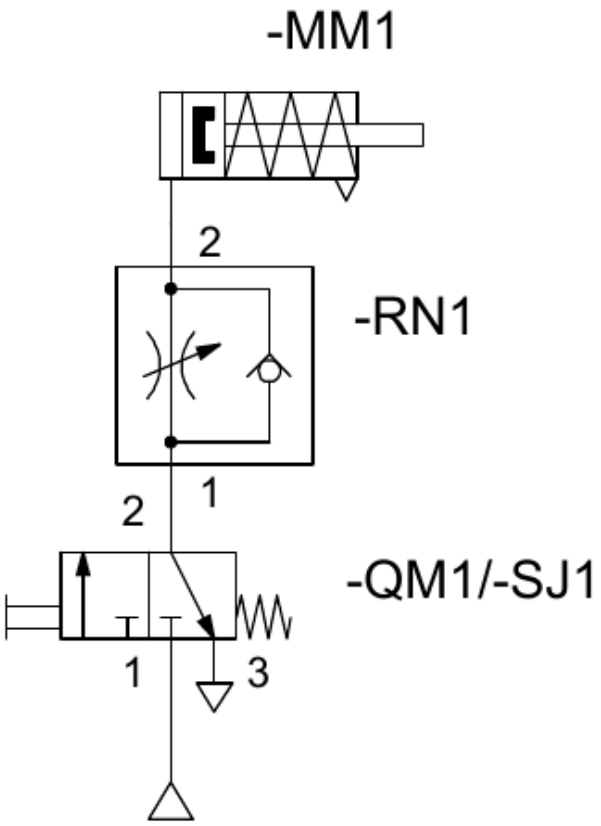
Same base circuit as Ex.1, with a plain throttle valve (-RN1) inserted between the valve and the cylinder. Since it restricts flow equally in both directions, it slows the cylinder on extend only (the retract stroke is driven by the spring and mostly vents past the valve, not through the throttle).



Exercise 3 — Single-acting cylinder, one-way (extend-only) speed control

Suggested filename: Ex3-SingleActing-OneWayFlowControl.circ

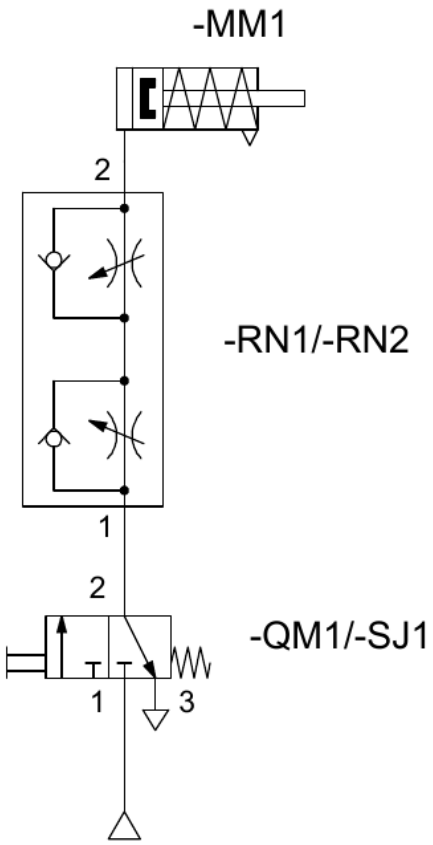
A one-way flow control valve (-RN1, throttle + check valve in one body) replaces the plain throttle. Oriented so the check valve allows free flow into the cylinder while throttling the air escaping it, this gives smooth, adjustable meter-out control on the extend stroke specifically, confirmed by live test — adjusting -RN1 changed the extend speed only.



Exercise 4 — Single-acting cylinder, independent bidirectional speed control

Suggested filename: Ex4-SingleActing-AntiParallel.circ

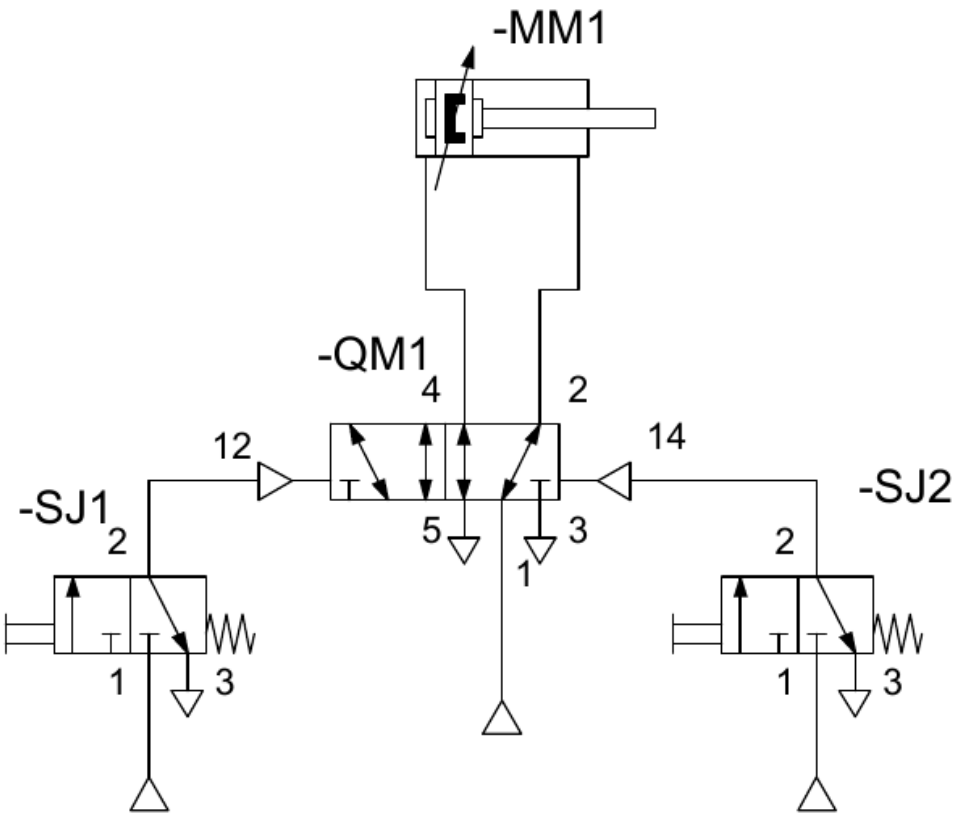
A double throttle-check valve (-RN1/-RN2, two one-way flow control units built into a single part, facing opposite directions) gives independent, adjustable control of both the extend and retract speeds. Confirmed via live test: adjusting each internal unit changed one direction of travel without affecting the other.



Exercise 5 — Double-acting cylinder, latching (SR) control

Suggested filename: Ex5-DoubleActing-Bistable.circ

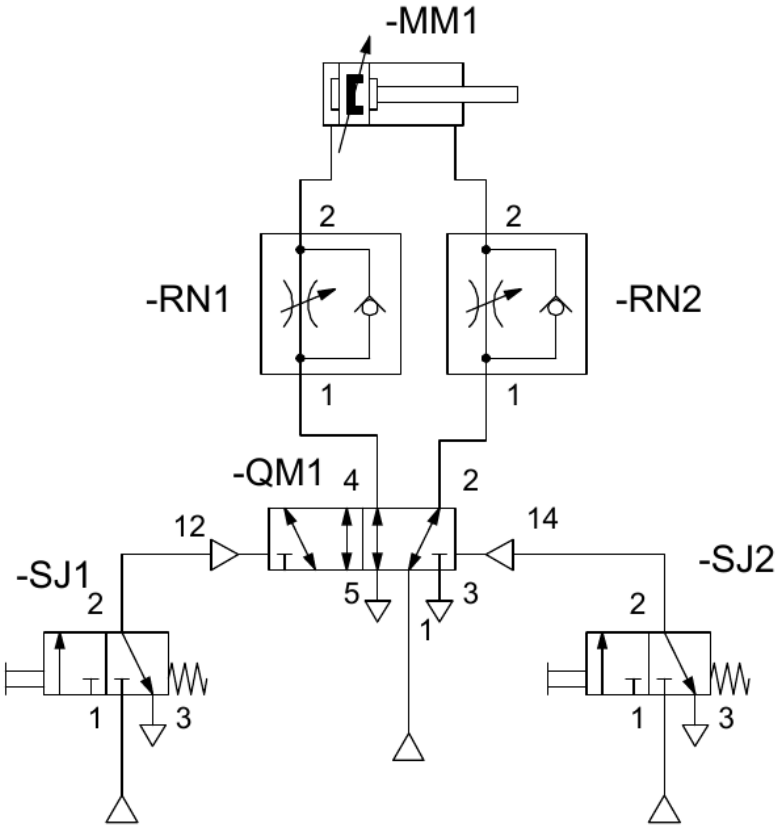
A double-acting cylinder (-MM1) is controlled by a bistable, double-pilot 5/2 valve (-QM1, no spring — pneumatically operated on both sides). A short pulse on -SJ1 shifts -QM1 to extend and it stays there; a short pulse on -SJ2 shifts it to retract and it stays there. This SR-latch behaviour is why the valve must be bistable rather than spring-return.



Exercise 6 — Double-acting cylinder, bidirectional speed control

Suggested filename: Ex6-DoubleActing-SpeedControl.circ

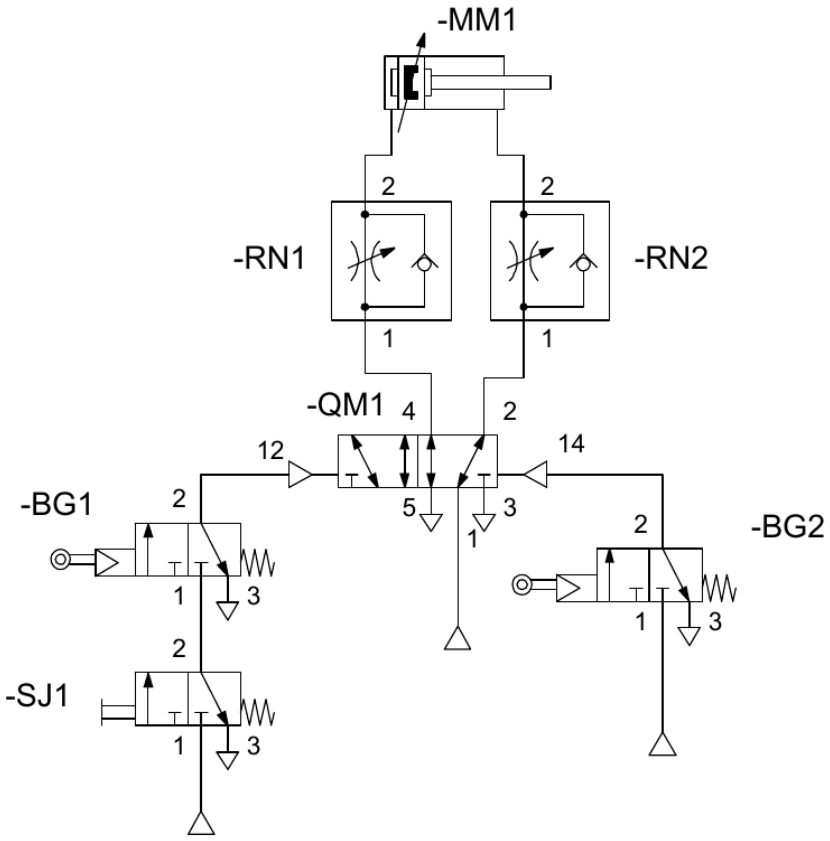
Same latching double-acting circuit as Ex.5, with a one-way flow control valve added on each working line (-RN1, -RN2), between -QM1 and the cylinder's two ports. Each is oriented for meter-out control — free flow in, throttled flow out — giving independently adjustable speed in both directions, confirmed by live test.



Exercise 7 — Sheet-metal bending device (first sequential exercise)

Suggested filename: Ex7-DoubleActing-SequentialAutoRetract.circ

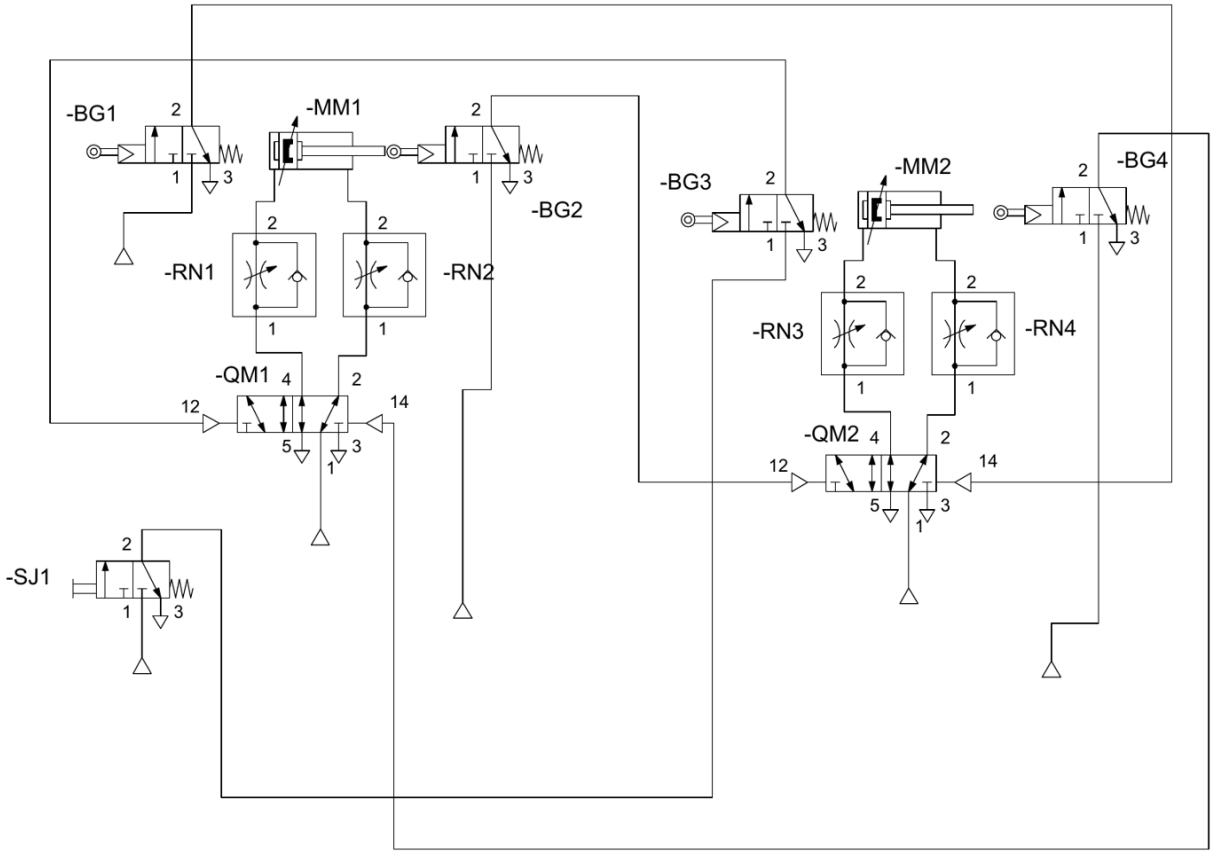
-MM1 extends (bending the sheet) when -SJ1 is pressed, but only if -MM1 is already fully retracted — enforced by wiring -SJ1 in series (AND logic) with limit switch -BG1, mounted at the retracted end, into pilot 12 of -QM1. Once -MM1 reaches full extension it strikes limit switch -BG2 (mounted at the extended end), which feeds pilot 14 of -QM1 directly, automatically retracting -MM1 with no second button press. Both directions keep the -RN1/-RN2 meter-out speed control from Ex.6.



Exercise 8 — Two-cylinder conveyor transfer

Suggested filename: Ex8-TwoCylinder-ConveyorTransfer.circ

Two double-acting cylinders (-MM1, -MM2), each with its own bistable 5/2 valve (-QM1, -QM2) and meter-out speed control (-RN1..-RN4). -SJ1 (interlocked in series with -BG3, confirming -MM2 is retracted) starts the sequence: -MM1 extends and holds. -MM1 reaching full extension (-BG2) triggers -MM2 to extend, pushing the part onto the second conveyor. -MM2 reaching full extension (-BG4) triggers -MM1 to retract. -MM1 reaching full retraction (-BG1) triggers -MM2 to retract, completing the cycle. Every limit switch fires the other cylinder's valve.



Exercise 9 — Two-cylinder drilling jig

Suggested filename: Ex9-TwoCylinder-DrillingJig.circ

Same building blocks as Ex.8. -SJ1 (interlocked in series with -BG1, confirming -MM1 is retracted) makes -MM1 extend, clamping the part. -MM1 reaching full extension (-BG2) triggers -MM2 to extend, driving the drill in. -MM2 reaching full extension (-BG4) feeds its own valve's other pilot directly, so -MM2 automatically retracts itself, withdrawing the drill — no cross-cylinder trigger needed for this step. -MM2 reaching full retraction (-BG3) then triggers -MM1 to retract, releasing the part. Restart requires -MM1 to be confirmed retracted again (the -BG1 interlock from step one).

